| | | | |
|---|---|---|---|
| **Fiscal Year:** | FY 2014 | **Task Last Updated:** | FY 08/20/2013 |
| **PI Name:** | Rabin, Bernard M. Ph.D. | | |
| **Project Title:** | Individual Differences in the Neurochemical and Behavioral Response to Exposure to Protons and HZE Particles | | |
| **Division Name:** | Human Research | | |
| **Program/Discipline:** | HUMAN RESEARCH | | |
| **Program/Discipline--Element/Subdiscipline:** | | | |
| **Joint Agency Name:** | | **TechPort:** | No |
| **Human Research Program Elements:** | (1) **SR**:Space Radiation | | |
| **Human Research Program Risks:** | (1) **BMed**:Risk of Adverse Cognitive or Behavioral Conditions and Psychiatric Disorders | | |
| **Space Biology Element:** | None | | |
| **Space Biology Cross-Element Discipline:** | None | | |
| **Space Biology Special Category:** | None | | |
| **PI Email:** | rabin@umbc.edu | **Fax:** | FY (410) 455-1055 |
| **PI Organization Type:** | UNIVERSITY | **Phone:** | (410) 952-1761 |
| **Organization Name:** | University of Maryland, Baltimore County | | |
| **PI Address 1:** | Department of Psychology | | |
| **PI Address 2:** | 1000 Hilltop Cir | | |
| **PI Web Page:** | | | |
| **City:** | Baltimore | **State:** | MD |
| **Zip Code:** | 21250-0001 | **Congressional District:** | 7 |
| **Comments:** | | | |
| **Project Type:** | Ground | **Solicitation / Funding Source:** | 2012 Space Radiobiology NNJ12ZSA001N |
| **Start Date:** | 10/23/2012 | **End Date:** | 10/22/2015 |
| **No. of Post Docs:** | | **No. of PhD Degrees:** | |
| **No. of PhD Candidates:** | | **No. of Master' Degrees:** | |
| **No. of Master's Candidates:** | | **No. of Bachelor's Degrees:** | 2 |
| **No. of Bachelor's Candidates:** | 4 | **Monitoring Center:** | NASA JSC |
| **Contact Monitor:** | Simonsen, Lisa | **Contact Phone:** | |
| **Contact Email:** | lisa.c.simonsen@nasa.gov | | |
| **Flight Program:** | | | |
| **Flight Assignment:** | | | |
| **Key Personnel Changes/Previous PI:** | None | | |
| **COI Name (Institution):** | Shukitt-Hale, Barbara ( US Department of Agriculture/Agricultural Research Service, Beltsville ) | | |
| **Grant/Contract No.:** | NNX13AB73G | | |
| **Performance Goal No.:** | | | |
| **Performance Goal Text:** | | | |
| **Task Description:** | Future missions in space (such as a mission to Mars or the establishment of a moon base) will involve long-term travel beyond the magnetic field of the Earth, subjecting astronauts to radiation hazards posed by solar flares and galactic cosmic rays. Cosmic rays are composed of protons, alpha particles, and particles of high energy and charge (HZE particles). Previous research has shown that exposure to HZE particles and protons can affect neural functioning and behavior. These changes may affect the ability of astronauts to successfully meet mission requirements or may produce degenerative changes in neural and behavioral functioning after the conclusion of the mission. The objectives of the proposed research program are: (1) to evaluate the effects of repeated subthreshold exposures to HZE particles and protons on cognitive and neuronal deficits and how they relate to the characteristics of the specific particle; (2) to determine the relationship between changes in performance and HZE particle- and proton-induced changes in oxidative stress, inflammation, signaling molecules and autophagy in key brain regions; and (3) to determine how individual | | |

| | |
|---|---|
| | characteristics, including age and sex, may influence the responsiveness of the individual to HZE particle irradiation. To achieve these objectives, we propose using behavioral and neurochemical measures that have been previously shown to be sensitive to exposure to HZE particles and protons and which may provide a basis for defining the risks of exposure to heavy particles on brain functioning and related behavior. |
| **Rationale for HRP Directed Research:** | |
| **Research Impact/Earth Benefits:** | |
| **Task Progress:** | Two sets of experiments were performed. The first set of experiments was designed to compare the effects of exposure to two energies of protons (150 and 1000 MeV/n) and two patterns of exposure (head-only, whole-body) on several measures of cognitive performance. Exposure to protons does not produce consistent decrements in cognitive performance across different laboratories or within the same laboratory at different times. This experiment was designed to determine whether differences in particle energy or in exposure pattern could account for the differences in results. The second series of experiments was designed to study the acute effects of exposure to 56Fe (600 MeV/n) and 16O (600 MeV/n) particles 12-24 hours post-irradiation on learning and memory using the novel object recognition task and to relate possible changes in cognitive performance to changes in central nervous system function. Both series of experiments are continuing.<br>Following exposure to two energies of protons (150 and 1000 MeV/n) and two exposure conditions (head-only and whole-body) the rats were shipped to UMBC to test for proton-induced changes in performance on a variety of cognitive tasks, including plus-maze performance (a measure of baseline anxiety), novel object recognition (learning and memory), and operant responding on an ascending fixed-ratio reinforcement schedule (a measure of motivation to work for reinforcement). There were no effects of exposure to protons on plus-maze or object recognition performance. Exposure to 150 MeV/n protons (head-only) seemed to result in improved performance compared to the non-irradiated control animals. However, this radiation-induced change in performance may reflect the fact that the performance of the control animals was extremely low. We are re-testing the animals at this time. Overall, the results following exposure to protons are concordant with previous research in showing that exposure to protons does not produce consistent decrements in cognitive performance.<br><br>The results of the experiments testing for the acute effects of exposure to HZE particles showed that exposing the rats to either 56Fe or 16O particles after conditioning (learning) but before testing, disrupted recognition memory: the rats spent equal amounts time with the familiar and novel objects, indicating that they did not recognize the familiar object. In contrast to the test of recognition memory, there was no effect of exposure to either particle on the ability of the subjects to learn the task when conditioning occurred 24 hr after exposure. When the rats were irradiated prior to conditioning, subjects were able to recognize the familiar object, as shown by the greater amount of time spent with the novel object in comparison to the familiar object. The results indicate that there are acute effects on cognitive performance. The disruption of memory by interposing exposure to HZE particles between learning (conditioning) and testing suggests that the irradiation interferes with the consolidation process: the process by which a memory is transferred from short-term to long-term memory. In the learning paradigm, in contrast, irradiation occurs before conditioning (learning) and therefore does not affect the consolidation process and recognition memory is preserved.<br><br>Thirty and sixty days following irradiation and the initial test, the remaining subjects were again tested for the effects of exposure to HZE particles on novel object recognition performance. There was no effect of exposure to either 56Fe or 16O particles on novel object recognition performance tested 30 days after radiation exposure. However, when tested again 60 days following irradiation, the rats exposed to 56Fe particles showed a disruption of recognition memory. The failure to observe an effect two months following exposure to 16O particles, may reflect the low dose (5 cGy) used in the present experiment. The observation of a cognitive deficit in rats tested 2 months after irradiation is generally consistent with previous research in which the initial testing occurs 1-3 months following exposure. The neurochemical analyses are currently underway so that it will be possible to correlate the behavioral changes with changes in central nervous system function. |
| **Bibliography Type:** | Description: (Last Updated: 05/08/2025) |
| **Abstracts for Journals and Proceedings** | Rabin BM, Shukitt-Hale B, Carrihill-Knoll KL. "Comparison of the effectiveness of 16O and 56Fe particles in disrupting cognitive performance in F-344 rats of different ages." 2013 NASA Human Research Program Investigators' Workshop, Galveston, TX, February 12-14, 2013.<br>2013 NASA Human Research Program Investigators' Workshop, Galveston, TX, February 12-14, 2013. , Feb-2013 |
| **Abstracts for Journals and Proceedings** | Poulose S, Bielinski DF, Carrihill-Knoll K, Rabin BM, Shukitt-Hale B. "Blueberries and strawberries activate neuronal housekeeping in critical brain regions of stress-induced young rats." Experimental Biology 2013 meeting, Boston, MA, April 20-24, 2013.<br>FASEB J. 2013 Apr;27(Meeting Abstract Supplement):637-39. See also http://www.fasebj.org/content/vol27/1_MeetingAbstracts for searching. , Apr-2013 |
| **Abstracts for Journals and Proceedings** | Wyrobek AJ, Lowe, XR, Rabin BM., Bhatnager, S, Rademaker P, Northen T, Witkowska E, Straume T, Snijders A. "Molecular defects in the choroid plexus and hippocampus of animals with neurocognitive deficits after exposure to space radiation." HITSRS2013--Heavy Ion in Therapy and Space Radiation Symposium 2013 and 6th International Workshop on Space Radiation Research, Chiba, Japan, May 15-18, 2013.<br>HITSRS2013--Heavy Ion in Therapy and Space Radiation Symposium 2013 and 6th International Workshop on Space Radiation Research, Chiba, Japan, May 15-18, 2013. Program and Abstracts, p. 114. , May-2013 |
| **Abstracts for Journals and Proceedings** | Poulose SM, Bielinski D, Carrihill-Knoll KL, Rabin BM, Shukitt-Hale B. "Neuronal stress following exposure to 56Fe particles and the effects of antioxidant-rich diets." HITSRS2013--Heavy Ion in Therapy and Space Radiation Symposium 2013 and 6th International Workshop on Space Radiation Research, Chiba, Japan, May 15-18, 2013.<br>HITSRS2013--Heavy Ion in Therapy and Space Radiation Symposium 2013 and 6th International Workshop on Space Radiation Research, Chiba, Japan, May 15-18, 2013. Program and Abstracts, p. 49. , May-2013 |

| | |
|---|---|
| **Abstracts for Journals and Proceedings** | Rabin BM, Shukitt-Hale B, Gomes S, Carrihill-Knoll L. "Cognitive effects of partial and whole body exposures to 16O particles." HITSRS2013--Heavy Ion in Therapy and Space Radiation Symposium 2013 and 6th International Workshop on Space Radiation Research, Chiba, Japan, May 15-18, 2013.<br>HITSRS2013--Heavy Ion in Therapy and Space Radiation Symposium 2013 and 6th International Workshop on Space Radiation Research, Chiba, Japan, May 15-18, 2013. Program and Abstracts, p. 111. , May-2013 |
| **Abstracts for Journals and Proceedings** | Rabin BM. "Effects of berry diets on tumorigenesis and on the disruption of cognitive performance produced by exposure to space radiation. " Presented at the 42nd Annual Meeting of the American Aging Association, Baltimore, MD, May 31-June 3, 2013.<br>42nd Annual Meeting of the American Aging Association, Baltimore, MD, May 31-June 3, 2013. Program and Abstracts, p. 35. , May-2013 |